

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To study different word embedding for representation of textual data in vectorized form	
Experiment No: 8	Date:	Enrolment No : 92301733022

Aim: To study different word embedding for representation of textual data in vectorized form

IDE: Google Colab

Theory:

Word Embedding is an approach for representing words and documents. Word Embedding or Word Vector is a numeric vector input that represents a word in a lower-dimensional space. It allows words with similar meanings to have a similar representation.

Word Embeddings are a method of extracting features out of text so that we can input those features into a machine learning model to work with text data. They try to preserve syntactical and semantic information. The methods such as Bag of Words (BOW), CountVectorizer and TFIDF rely on the word count in a sentence but do not save any syntactical or semantic information. In these algorithms, the size of the vector is the number of elements in the vocabulary. We can get a sparse matrix if most of the elements are zero. Large input vectors will mean a huge number of weights which will result in high computation required for training. Word Embeddings give a solution to these problems.

Need for Word Embedding?

- To reduce dimensionality
- To use a word to predict the words around it.
- Inter-word semantics must be captured.

How are Word Embeddings used?

- They are used as input to machine learning models. Take the words —> Give their numeric representation —> Use in training or inference.
- To represent or visualize any underlying patterns of usage in the corpus that was used to train them.

1. Bag of Words

Bag of Words (BOW) is an algorithm that counts how many times a word appears in a document. Those word counts allow us to compare documents and gauge their similarities for applications like search, document classification, and topic modeling.

2. Tf-Idf

Tf-Idf is shorthand for term frequency-inverse document frequency. So, two things: term frequency and inverse document frequency. Term frequency (TF) is basically the output of the BoW model. For a specific document, it determines how important a word is by looking at how frequently it appears in the document. Term frequency measures the importance of the word. If a word appears a lot of times, then the word must be important. For example, if our document is “I am a cat lover. I have a cat named Steve. I feed a cat outside my room regularly,” we see that the words with the highest frequency are I, a, and cat. This agrees with our intuition that high term frequency = higher importance.

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$TF(t) = (\text{Number of times term } t \text{ appears in a document}) / (\text{Total number of terms in the document})$

IDF used to calculate the weight of rare words across all documents. The words that occur rarely in the corpus have a high IDF score. However, it is known that certain terms, such as “I”, “a” may appear a lot of times but have little importance. Thus we need to weigh down the frequent terms while scaling up the rare ones

$IDF(t) = \log_e(\text{Total number of documents} / \text{Number of documents with term } t \text{ in it})$

Pre Lab Exercise:



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University

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Department of Information and Communication Technology

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[1]

✓ 0s

```
corpus = [  
    "I love machine learning",  
    "Machine learning is amazing",  
    "I love coding in python"  
]
```

[2]

✓ 2s



```
from sklearn.feature_extraction.text import CountVectorizer
```

```
# Initialize vectorizer
```

```
bow = CountVectorizer()
```

```
# Fit and transform
```

```
X_bow = bow.fit_transform(corpus)
```

```
# Convert to array
```

```
bow_array = X_bow.toarray()
```

```
# Get feature names
```

```
features = bow.get_feature_names_out()
```

```
# Print results
```

```
print("Vocabulary:", features)
```

```
print("\nBoW Matrix:\n", bow_array)
```

```
... Vocabulary: ['amazing' 'coding' 'in' 'is' 'learning' 'love' 'machine' 'python']
```

```
BoW Matrix:
```

```
[[0 0 0 0 1 1 1 0]
```

```
[1 0 0 1 1 0 1 0]
```

```
[0 1 1 0 0 1 0 1]]
```

[3]

✓ 0s

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
# Use TF only (disable IDF)
```

```
tf = TfidfVectorizer(use_idf=False, norm='l1')
```

```
# Fit and transform
```

```
X_tf = tf.fit_transform(corpus)
```

```
# Convert to array
```

```
tf_array = X_tf.toarray()
```

```
# Get feature names
```

```
features_tf = tf.get_feature_names_out()
```

```
# Print results
```

```
print("Vocabulary:", features_tf)
```

```
print("\nTF Matrix:\n", tf_array)
```

```
Vocabulary: ['amazing' 'coding' 'in' 'is' 'learning' 'love' 'machine' 'python']
```

```
TF Matrix:
```

```
[[0.          0.          0.          0.          0.33333333 0.33333333
```

```
 0.33333333 0.          ]
```

```
[0.25         0.          0.          0.25         0.25         0.
```

```
 0.25         0.          ]
```

```
[0.          0.25         0.25         0.          0.          0.25
```

```
 0.          0.25         ]]
```

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Program (Code):

sentences = ["I feel proud to be a citizen of India because of its rich culture, traditions, and history.", "It is known for its festivals, traditions, and historical monuments.", "India is the world's largest democracy.",

"Technology makes our work easier and faster.", "It helps us communicate and learn better.", "Modern life depends heavily on technology.",

"Sports help people stay healthy and active.", "They teach teamwork, discipline, and confidence.", "in my free time i would love to play cricket."

```
]
#importing the libraries
import numpy as np
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer #TF-
Bow from sklearn.feature_extraction.text import TfidfVectorizer
#TFIDF from sklearn.metrics.pairwise import cosine_similarity
```

```
#Embeddings using TF_Bow approach
tfidf_vectorizer = TfidfVectorizer()
tfidf_matrix =
tfidf_vectorizer.fit_transform(sentences) tfidf_matrix
tfidf_similarity=cosine_similarity(tfidf_matrix)
tfidf_similarity
```

```
#finding out the document similarity
def
doc_sim(similarity_matrix,sentences)
: for i in range(len(sentences)):
sims=similarity_matrix[i].copy()
sims[i]=-1
sims=sims.max()
top_index=np.argmax(sims)
print(sentences[i],"--
>",sentences[top_index])
doc_sim(tfidf_similarity,sentences)
```

Results:

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```

** array([[1.          , 0.15310398, 0.07874408, 0.02841468, 0.02841468,
          0.          , 0.02775765, 0.03027674, 0.06214335],
        [0.15310398, 1.          , 0.10744065, 0.03876979, 0.14108832,
          0.          , 0.03787332, 0.04131044, 0.          ],
        [0.07874408, 0.10744065, 1.          , 0.          , 0.          ,
          0.          , 0.          , 0.          , 0.          ],
        [0.02841468, 0.03876979, 0.          , 1.          , 0.04517405,
          0.12200786, 0.0441295 , 0.04813438, 0.          ],
        [0.02841468, 0.14108832, 0.          , 0.04517405, 1.          ,
          0.          , 0.0441295 , 0.04813438, 0.          ],
        [0.          , 0.          , 0.          , 0.12200786, 0.          ,
          1.          , 0.          , 0.          , 0.          ],
        [0.02775765, 0.03787332, 0.          , 0.0441295 , 0.0441295 ,
          0.          , 1.          , 0.04702137, 0.          ],
        [0.03027674, 0.04131044, 0.          , 0.04813438, 0.04813438,
          0.          , 0.04702137, 1.          , 0.          ],
        [0.06214335, 0.          , 0.          , 0.          , 0.          ,
          0.          , 0.          , 0.          , 1.          ]])

```

I feel proud to be a citizen of India because of its rich culture, traditions, and history. --> I feel proud to be a citizen of India because of its rich culture, traditions, and history.
It is known for its festivals, traditions, and historical monuments. --> I feel proud to be a citizen of India because of its rich culture, traditions, and history.
India is the world's largest democracy. --> I feel proud to be a citizen of India because of its rich culture, traditions, and history.
Technology makes our work easier and faster. --> I feel proud to be a citizen of India because of its rich culture, traditions, and history.
It helps us communicate and learn better. --> I feel proud to be a citizen of India because of its rich culture, traditions, and history.
Modern life depends heavily on technology. --> I feel proud to be a citizen of India because of its rich culture, traditions, and history.
Sports help people stay healthy and active. --> I feel proud to be a citizen of India because of its rich culture, traditions, and history.
They teach teamwork, discipline, and confidence. --> I feel proud to be a citizen of India because of its rich culture, traditions, and history.
In my free time i would love to play cricket. --> I feel proud to be a citizen of India because of its rich culture, traditions, and history.

Observation:

- BoW, TF, and TF-IDF algorithms were used to transform three documents into vector form, and cosine similarity was used to calculate similarity.
- Because BoW vectorization bases similarity on raw word counts, documents that share common words like "machine" and "learning" exhibit increased similarity even when their meanings are not closely connected.
- By taking into account the proportion of words rather than raw counts, normalization enhances the comparison in TF vectorization and produces slightly more balanced similarity values.
- Rare and significant words are given more weight in TF-IDF vectorization, while common words are given less weight, producing more precise and significant similarity results.
- It was shown that while unrelated papers displayed lower similarity across all techniques, documents with similar contexts had greater similarity scores.

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Post Lab Exercise:

```
[4]
✓ Os documents = [
    "I love machine learning",
    "Machine learning is amazing",
    "I enjoy coding in python"
]

[5]
✓ Os from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# BoW
bow = CountVectorizer()
X_bow = bow.fit_transform(documents)

# Similarity
bow_similarity = cosine_similarity(X_bow)

print("BoW Similarity Matrix:\n", bow_similarity)

BoW Similarity Matrix:
[[1. 0.57735027 0. ]
 [0.57735027 1. 0. ]
 [0. 0. 1. ]]

[6]
✓ Os from sklearn.feature_extraction.text import TfidfVectorizer

# TF (no IDF)
tf = TfidfVectorizer(use_idf=False, norm='l2')
X_tf = tf.fit_transform(documents)

# Similarity
tf_similarity = cosine_similarity(X_tf)

print("TF Similarity Matrix:\n", tf_similarity)

TF Similarity Matrix:
[[1. 0.57735027 0. ]
 [0.57735027 1. 0. ]
 [0. 0. 1. ]]

[7]
✓ Os # TF-IDF
tfidf = TfidfVectorizer()
X_tfidf = tfidf.fit_transform(documents)

# Similarity
tfidf_similarity = cosine_similarity(X_tfidf)

print("TF-IDF Similarity Matrix:\n", tfidf_similarity)

TF-IDF Similarity Matrix:
[[1. 0.44333251 0. ]
 [0.44333251 1. 0. ]
 [0. 0. 1. ]]
```

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- Discuss the results obtained in each scenario.

BoW Vectorization (a)

- Raw word counts are used by BoW to measure similarity.
- There is a lot of similarities in documents that use terms like "machine" and "learning."
- The significance of words is not taken into account.
- The majority of similarity scores are common words.
- Less precise when comparing semantics.

TF Vectorization (b).

- TF improves comparison by standardizing word counts.
- Because sentences are lengthier, prejudice is lessened.
- Similarity is more evenly distributed than BoW.
- Regardless of their significance, frequently used terms are nonetheless valued.
- A minor improvement over BoW.

TF-IDF Vectorization (c).

- TF-IDF assigns more weight to significant (uncommon) terms.
- Words like "I" and "is" become less significant.
- Generates results that are more meaningfully similar.
- The most effective of the three techniques for text similarity.
- Documents the significance of words.